

Notes: Mian and Sufi (Econometrica, 2014)
What Explains the 2007–2009 Drop in Employment?

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1 Big picture

- **Research question.** What explains the sharp fall in U.S. employment during the Great Recession? More specifically, did the deterioration of household balance sheets through falling housing net worth cause local employment losses?
- **Main idea.** When local households lose housing net worth, they cut spending. Industries that sell locally, such as restaurants and retail stores, should then lose more jobs in the counties where housing net worth falls most.
- **Key distinction.** The paper separates **non-tradable** employment, which depends on local demand, from **tradable** employment, which depends on national or global demand. This split is the core empirical strategy.
- **Main finding.** Counties with larger housing net worth losses from 2006 to 2009 have much larger declines in non-tradable employment from 2007 to 2009. The relationship appears under two independent industry-classification methods.
- **Magnitude.** A 10 percentage point decline in housing net worth is associated with a sizable decline in non-tradable employment. The IV estimate in the paper implies about a 3.7 percentage point decline in restaurant-and-retail non-tradable employment.
- **Ruling out alternatives.** The result is not explained by exposure to construction, by supply-side shocks to particular industries, by policy uncertainty, or by contemporaneous local credit-supply tightening.
- **Adjustment failure.** The paper finds little evidence that harder-hit counties offset local demand shocks through lower wages, in-migration outflows, labor force adjustment, or expansion of tradable employment.
- **Economic takeaway.** The Great Recession employment collapse was partly a local aggregate-demand shock caused by household balance sheet deterioration. Weak labor-market adjustment allowed local demand losses to translate into real job losses.

2 Incremental contribution of the paper

- **From spending effects to employment effects.** Earlier evidence shows that household balance sheet deterioration reduces consumption. Mian and Sufi ask whether this spending decline also caused employment losses.
- **Tradable versus non-tradable identification.** The paper uses the idea that local demand shocks should affect non-tradable jobs much more than tradable jobs. This provides a way to distinguish local demand from broad national industry shocks.
- **Two independent industry classifications.** The paper classifies industries using both a direct trade/retail definition and a geographical-concentration definition. Similar results across both definitions strengthen the interpretation.
- **County-level balance sheet shock.** The paper constructs a county-level housing net worth shock using household leverage, housing values, and house-price declines.

- **Broad robustness against competing channels.** The authors test construction exposure, industry composition, Saiz housing supply elasticity as an instrument, local banking exposure, establishment size, and multiple labor-market adjustment margins.
- **Simple model to discipline interpretation.** The island model clarifies what should happen under flexible wages and prices: non-tradable job losses in hard-hit counties should be offset by tradable employment gains. The absence of this offset is central.
- **New tradability measure.** The geographical concentration index gives a continuous measure of tradability that can be reused in other regional-shock research.
- **Macro implication from micro variation.** The paper uses cross-county variation to provide evidence on a macro question: why aggregate employment fell so much during 2007–2009.

3 Data and measurement

3.1 County employment and wage data

The paper builds a county-level panel that links local employment, payroll, wages, household balance sheets, and demographic characteristics.

- **Employment and payroll.** County-by-four-digit-industry employment and payroll come from the Census County Business Patterns (CBP). CBP employment is recorded in March of each year.
- **Wages.** Payroll-based average wages come from CBP. Hourly wages come from the American Community Survey (ACS), which is important because payroll per employee can change if hours change.
- **County sample.** The main sample contains 944 counties for which the housing net worth shock can be constructed. These counties represent about 80 percent of total U.S. population.
- **Timing.** The main employment outcomes measure growth from 2007 to 2009, while the housing net worth shock is measured from 2006 to 2009.

Interpretation. The data allow the authors to compare counties that were exposed to different household balance sheet shocks while observing employment by detailed industry.

3.2 Housing net worth shock

Household net worth in county i is conceptually written as

$$NW_{i,t} = S_{i,t} + B_{i,t} + H_{i,t} - D_{i,t},$$

where S is stock holdings, B is bonds and deposits, H is housing value, and D is household debt.

The key shock, denoted here by ΔHNW_i , is the housing-driven change in household net worth between 2006 and 2009. Intuitively, it is larger in magnitude where:

- house prices fall more;

- households entered the recession with more housing exposure;
- household leverage made a given house-price fall more damaging to net worth.

Negative values mean larger household balance sheet losses. The paper treats this shock as the local demand shock induced by the collapse in housing wealth and collateral values.

3.3 Classifying industries

The paper places four-digit industries into non-tradable, tradable, construction, or other categories.

Classification 1: retail, restaurants, and U.S.–world trade.

- Non-tradable industries are retail and restaurants.
- Tradable industries are industries with sufficiently large imports plus exports per worker or total trade volume.
- Construction, real estate, and land-development industries are separated from both groups.

Classification 2: geographical concentration.

- Industries serving national or global demand should be geographically concentrated because production can specialize by location.
- Industries serving local demand should be geographically dispersed because every county needs them.
- The paper constructs an industry-level geographical Herfindahl index using the distribution of industry employment across counties.
- The top quartile of concentration is classified as tradable; the bottom quartile is classified as non-tradable.

Interpretation. The first classification is intuitive and transparent. The second is independent and more general, which makes it a useful robustness check.

3.4 Main empirical objects

The paper studies several outcomes:

- non-tradable employment growth;
- tradable employment growth;
- construction employment growth;
- wage growth, using both CBP payroll data and ACS hourly wage data;
- population growth, in-migration, and labor force growth;
- county-industry employment growth interacted with industry tradability.

4 Methodology and empirical specifications

4.1 Non-tradable employment response

The baseline regression is

$$\Delta \log E_i^N = \alpha + \eta \Delta HNW_i + \varepsilon_i,$$

where $\Delta \log E_i^N$ is county i 's non-tradable employment growth from 2007 to 2009 and ΔHNW_i is the housing net worth shock.

Because ΔHNW_i is negative in counties with larger housing losses, a positive coefficient η means that larger housing net worth losses predict larger employment declines.

The paper estimates this equation under both non-tradable definitions. Regressions are weighted by the number of households, and standard errors are clustered at the state level. The paper also reports spatially robust standard errors in the main non-tradable table.

4.2 Supply-side and construction controls

A key concern is that counties with large housing losses might also be exposed to industries that were hit by unrelated supply-side shocks.

The paper addresses this in several ways:

- controls for 23 two-digit industry employment shares in 2006;
- instruments the housing net worth shock with Saiz housing supply elasticity;
- controls for county demographics and pre-recession characteristics;
- interacts the housing net worth shock with the county construction employment share;
- directly controls for construction employment growth.

Interpretation. The housing net worth coefficient remains positive and economically important, so the result is not simply a construction-sector story.

4.3 Credit supply tests

Another alternative is that falling house prices reduced local business credit supply, causing firms to lay off workers. The paper tests this by using establishment size and banking geography.

If local credit tightening drives the result, the effect should be stronger among small establishments that are more credit constrained. Instead, Table IV shows stronger effects among larger establishments. The paper also compares counties served mainly by national banks versus local banks and finds similar patterns.

Interpretation. These tests do not rule out all credit-supply effects in the economy, but they make it unlikely that local credit supply tightening is the main explanation for the non-tradable result.

4.4 Simple island model

The model has many counties or “islands.” Each county produces tradable and non-tradable goods. Non-tradable goods must be consumed locally, while tradable goods can be sold nationally.

Local demand is hit by shocks δ_c . With full nominal or real rigidity, job losses in county c are

$$Y_c = \alpha\delta_c + (1 - \alpha)\bar{\delta},$$

where α is the non-tradable expenditure share and $\bar{\delta}$ is the aggregate demand shock.

The formula captures the key empirical distinction:

- non-tradable employment depends on the local shock δ_c ;
- tradable employment depends on aggregate demand, not the county-specific shock;
- under flexible wages and prices, tradable employment should expand in harder-hit counties and offset the non-tradable decline.

4.5 Testing labor-market adjustment

The paper tests the adjustment mechanisms implied by the model:

- Does tradable employment rise in counties with larger housing net worth declines?
- Do wages fall more in those counties?
- Do people move away from those counties?
- Does the labor force shrink more in those counties?

The answer is mostly no. Tradable employment does not rise in hard-hit counties, wages do not adjust strongly downward, and mobility responses are weak.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies whether counties with larger housing net worth losses experienced larger employment losses in industries tied to local demand, relative to industries tied to national or global demand.

The strongest causal interpretation is that the housing net worth shock reduced local demand and thereby reduced non-tradable employment. The identification is based on cross-county variation, industry tradability, robustness controls, and an instrument based on housing supply elasticity.

5.2 Why non-tradable employment is central

The housing net worth channel predicts local spending declines. Local spending should affect restaurants, retail stores, and other geographically dispersed services much more directly than industries selling to national or global markets.

This makes non-tradable employment the cleanest outcome for detecting local demand effects. Tradable employment is then a placebo-like comparison and a test of labor-market adjustment.

5.3 Interpreting the Saiz instrument

Housing supply elasticity affects how much house prices rise during the boom and fall during the bust. Less elastic areas should see larger price movements and therefore larger housing net worth shocks.

The instrument helps address reverse causality and omitted variables tied to local housing price declines. The exclusion restriction is that housing supply elasticity affects non-tradable employment during 2007–2009 through the housing net worth channel rather than through other county shocks.

5.4 Credit supply versus demand

A local credit supply shock could also reduce employment. The paper’s evidence cuts against this view because the non-tradable effect is not concentrated in smaller establishments and does not vary in the way one would expect if local banks’ balance sheets were the key channel.

The demand interpretation is also supported by the absence of a similar relationship in tradable employment. If local firm credit constraints were the common driver, tradable employers should also be affected.

5.5 Why tradable non-response matters

Under flexible wages and frictionless reallocation, workers released from non-tradable jobs in hard-hit counties should be absorbed by the tradable sector after local wages fall. The paper finds no such offset.

This non-response suggests that the local demand shock translates into real employment losses because wages, prices, search, skill matching, or mobility do not adjust fast enough.

5.6 Aggregate interpretation

The paper uses local variation to study a national recession. The local estimates do not mechanically sum to the aggregate employment loss because spillovers through trade and general equilibrium effects matter. Still, the pattern is consistent with the view that household balance sheet deterioration was an important aggregate-demand force in the Great Recession.

6 Tables and figures

6.1 Table I: Industry classification

Read:

- Table I lists the top industries classified as non-tradable and tradable under the two classification schemes.
- Panel A uses the direct retail/restaurant and U.S.–world-trade method.
- Panel B uses geographical concentration to identify industries tied to broader demand and industries tied to local demand.
- The two methods are deliberately different. Their agreement in the main results reduces concern that the findings are an artifact of a particular industry definition.

Economic message: the paper’s identification depends on separating local-demand industries from broad-demand industries; Table I shows how that separation is constructed.

Non-Tradable Industries				Tradable Industries							
NAICS	Industry Name	NT?	NAICS	Industry Name	T?	NAICS	Industry Name	T?			
Panel A: Industry classification based on retail, restaurants, and US—world trade						Panel B: Industry classification based on geographical concentration of industries					
7221	Full-service restaurants	1	3261	Plastics product manufacturing	0	4442	Lawn and garden equipment stores	0	5252	Securities and commodity exchanges	0
7222	Limited-service eating places	1	3231	Printing and related support activities	0	4245	Farm product raw material wholesalers	0	4861	Pipeline transportation of crude oil	0
4451	Grocery stores	1	3363	Motor vehicle parts manufacturing	0	4471	Gasoline stations	1	3152	Cut and sew apparel manufacturing	1
4521	Department stores	1	3116	Animal slaughtering and processing	0	2123	Nonmetallic mineral mining & quarrying	0	5121	Motion picture and video industries	0
4529	Other general merchandise stores	1	3364	Aerospace product & parts manufacturing	1	4529	Other general merchandise stores	1	7114	Agents for artists, entertainers, etc.	0
4481	Clothing stores	0	3327	Machine shops; screw cut & bolt manuf.	0	7212	RV parks and recreational camps	0	4831	Deep sea / great lakes water transportation	0
4461	Health and personal care stores	1	3345	Navigation & control instruments manuf.	0	3211	Sawmills and wood preservation	0	5152	Cable and other subscription programming	0
4471	Gasoline stations	1	3344	Semiconductor and other electronic manuf.	1	4531	Florists	1	5122	Sound recording industries	0
7223	Special food services	0	3399	Other miscellaneous manufacturing	1	8122	Death care services	0	3122	Tobacco manufacturing	1
4511	Sporting goods hobby and music stores	1	5112	Software publishers	1	5323	General rental centers	0	7115	Independent artists, writers and performers	0
7224	Drinking places (alcoholic beverages)	0	3391	Medical equipment and supplies manuf.	0	4543	Direct selling establishments	0	3365	Railroad rolling stock manufacturing	1
4532	Office supplies stationery and gift stores	1	3222	Converted paper product manufacturing	0	4441	Building material and supplies dealers	0	4879	Scenic and sightseeing transportation other	0
4539	Other miscellaneous store retailers	1	3118	Bakeries and tortilla manufacturing	0	4412	Other motor vehicle dealers	1	7131	Amusement parks and arcades	0
4482	Shoe stores	0	3339	Other general purpose machinery manuf.	0	6251	Nursing care facilities	0	4872	Sightseeing transportation water	0
4512	Book, periodical, and music stores	0	3329	Other fabricated metal product manuf.	0	4413	Automotive accessories and tire stores	1	5231	Securities and commodity intermediation	0
4452	Specialty food stores	0	3254	Pharmaceutical and medicine manuf.	0	1133	Logging	0	5181	Internet Service Providers	0
4483	Jewelry luggage and leather goods stores	1	3331	Agriculture and mining machinery manuf.	0	4842	Specialized freight trucking	0	2122	Metal ore mining	1
4453	Beer wine and liquor stores	1	3361	Motor vehicle manufacturing	1	3273	Cement and concrete manufacturing	0	4883	Support activities for water transportation	0
4533	Used merchandise stores	1	3251	Basic chemical manufacturing	1	3219	Other wood product manufacturing	0	4243	Apparel piece goods and wholesalers	0
4531	Florists	1	3114	Fruit & vegetable preserving & manuf.	0	6232	Mental health & substance abuse facilities	0	4889	Other support activities for transportation	0

Table I, Panel A: Industry classification based on retail, restaurants, and U.S.–world trade.

Table I, Panel B: Industry classification based on geographical concentration.

6.2 Table II and Figure 1: Summary statistics and the main non-tradable pattern

Read Table II:

- The average weighted housing net worth shock from 2006 to 2009 is about -9.5 percent.
- Total employment falls by about 5.3 percent in the weighted sample.
- Construction employment falls the most, but tradable and non-tradable employment also decline.
- The sample contains 944 counties for most county-level outcomes.

Read Figure 1:

- The figure plots non-tradable employment growth against the change in housing net worth.
- Both panels show a strong positive slope: counties with larger housing net worth losses have larger non-tradable job losses.
- The relationship appears with both definitions of non-tradable employment.

Economic message: the central visual fact is that local balance sheet deterioration is tightly linked to local non-tradable employment losses.

TABLE II
SUMMARY STATISTICS^a

	N	Mean	SD	10th	90th	Weighted Mean	Weighted SD
Housing net worth shock, 2006 to 2009	944	-0.065	0.085	-0.172	0.003	-0.095	0.100
Number of households, 2000	944	98,197	187,506	12,841	237,783	455,860	666,240
Labor force growth, 2007 to 2009	944	0.014	0.030	-0.018	0.050	0.014	0.025
Total employment, 2007	944	110,725	235,669	9,652	267,278	543,470	809,861
Employment growth, 2007 to 2009	944	-0.052	0.066	-0.123	0.021	-0.053	0.047
Average wage, 2007	944	7.338	2.414	5.234	9.985	9.727	3.790
Average wage growth, 2007 to 2009	944	0.028	0.071	-0.044	0.100	0.026	0.056
Housing supply elasticity (Saiz)	540	2.204	1.117	0.943	3.589	1.718	0.990
Non-tradable employment growth, 2007 to 2009	944	-0.029	0.086	-0.110	0.063	-0.040	0.061
Food industry employment growth, 2007 to 2009	944	-0.012	0.090	-0.093	0.089	-0.021	0.063
Tradable employment growth, 2007 to 2009	944	-0.115	0.192	-0.337	0.062	-0.116	0.136
Construction employment growth, 2007 to 2009	944	-0.163	0.164	-0.368	0.023	-0.161	0.136
Other employment growth, 2007 to 2009	944	-0.021	0.082	-0.103	0.070	-0.026	0.052
Industry geographical Herfindahl, 2007	294	0.016	0.023	0.0034	0.0338	0.0083	0.011
Hourly wage, 2007	944	18.978	3.447	15.484	23.354	21.086	3.692
Hourly wage, 10th percentile, 2007	944	5.801	0.830	4.834	7.000	6.241	0.774
Hourly wage, 25th percentile, 2007	944	9.052	1.450	7.500	10.955	9.808	1.464
Hourly wage, median, 2007	944	22.975	4.697	18.269	29.101	25.683	5.109
Hourly wage, 75th percentile, 2007	944	34.714	7.487	27.404	44.535	39.478	8.658
Hourly wage, 90th percentile, 2007	944	14.494	2.710	11.731	18.229	15.984	2.880
Wage growth, 2007 to 2009	943	0.012	0.089	-0.099	0.124	0.011	0.066
Wage growth, 10th percentile, 2007 to 2009	943	0.053	0.064	-0.022	0.137	0.048	0.049
Wage growth, 25th percentile, 2007 to 2009	943	0.058	0.055	-0.006	0.134	0.051	0.041
Wage growth, median, 2007 to 2009	943	0.050	0.068	-0.030	0.136	0.040	0.048
Wage growth, 75th percentile, 2007 to 2009	943	0.066	0.057	-0.001	0.137	0.056	0.042
Wage growth, 90th percentile, 2007 to 2009	943	0.039	0.057	-0.031	0.107	0.032	0.039

^a This table presents summary statistics for the county-level data used in the analysis. Employment data are from the Census County Business Patterns, wage data are from the American Community Survey, debt data are from Equifax, and income data are from the IRS. The last two columns are weighted by the number of households in the county as of 2000, except industry-level Herfindahl, which is weighted by an industry's 2007 total employment. The data are restricted to the 944 counties for which the housing net worth shock variable can be constructed. These counties represent 80% of total U.S. population.

Table II: Summary statistics.

6.3 Tables III and IV: Main regressions and credit-supply tests

Read Table III:

- The dependent variable is non-tradable employment growth from 2007 to 2009.
- The housing net worth coefficient is positive and significant under both non-tradable definitions.
- The result survives controls for two-digit industry shares, IV estimation using housing supply elasticity, demographics, construction share interactions, and construction employment growth.
- The IV estimates are often larger than the OLS estimates.

Read Table IV:

- The table asks whether non-tradable job losses are really driven by credit-supply tightening.
- If credit constraints drive the result, the effect should be strongest among small establishments.
- Instead, the effect is strongest among larger establishments.
- The result also appears in counties served by national banks and local banks.

Economic message: the main non-tradable employment result is robust to construction and industry

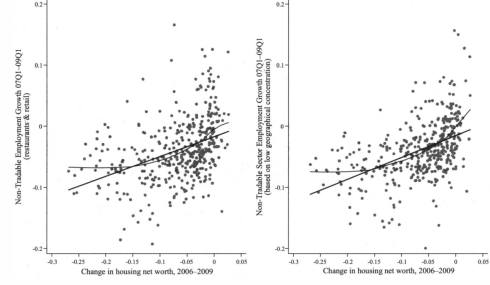


FIGURE 1.—Non-tradable employment and the housing net worth shock. This figure presents scatter-plots of county-level non-tradable employment growth from 2007Q1 to 2009Q1 against the change in housing net worth from 2006 to 2009. The left panel defines industries in restaurant and retail sector as non-tradable, and the right panel defines industries as non-tradable if they are geographically dispersed throughout the United States. The sample includes counties with more than 50,000 households. The thin black line in the left panel is the non-parametric plot of non-tradable employment growth against change in housing net worth.

Figure 1: Non-tradable employment and the housing net worth shock.

controls, and the credit-supply evidence points away from local credit tightening as the main mechanism.

Non-Tradable Definition Used:	Employment Growth, Non-Tradable Industries, 2007–2009									
	Rest. & Retail (1)	Geog. Consum. (2)	Rest. & Retail (3)	Geog. Consum. (4)	Rest. & Retail (5)	Geog. Consum. (6)	Rest. & Retail (7)	Rest. & Retail (8)	Geog. Consum. (9)	Rest. & Retail (10)
Δ Housing Net Worth, 2006–2009	0.190** (0.042) [0.022]	0.199** (0.049) [0.017]	0.174** (0.043) [0.021]	0.166** (0.046) [0.016]	0.374** (0.132) [0.081]	0.208* (0.086) [0.067]	0.489** (0.127) [0.072]	0.440** (0.140) [0.057]	0.212* (0.091) [0.057]	0.133** (0.036) [0.022]
Δ HNW * (Construction Share 07)							–1.99* (0.856)	–0.325 (0.561)		
Construction Share 07							–0.082 (0.158)	–0.183 (0.126)		
Δ Construction Employment, 2007–2009									0.079** (0.027)	
Constant	–0.022** (0.007)	–0.021** (0.007)	0.176 (0.443)	0.070 (0.286)	0.445 (0.536)	1.233** (0.438)	–0.102 (0.57)	0.254 (0.428)	0.072 (0.290)	0.162 (0.430)
Specification	OLS	OLS	OLS	OLS	IV	IV	IV	OLS	OLS	OLS
Industry controls?	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Other controls?	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
N	944	944	944	944	540	540	540	944	944	944
R ²	0.096	0.156	0.175	0.236	0.158	0.275	0.144	0.188	0.239	0.194

*This table reports regression estimates of non-tradable employment growth from 2007 to 2009 on the housing net worth shock at the county level. Regressions are weighted by the total number of households in a county. The IV specification uses housing supply elasticity as an instrument for the housing net worth shock. Standard errors in parentheses are clustered at the state level, and in square brackets are adjusted for spatial correlation with correlation proportional to the inverse of the distance between any two counties. The industry controls are the 2006 employment share in a county for 23 two-digit industries: Agriculture, Mining, Utilities, Construction, Manufacturing (3 two-digit industries), Wholesale Trade, Retail Trade (2 two-digit industries), Transportation (2 two-digit industries), Information, Finance, Real Estate, Professional Services, Management, Administrative Services, Education, Health Care, Entertainment, Accommodation and Food Services, Other Services. The other controls include pre-recession percentage white, median household income, percentage owner-occupied, percentage with less than high school diploma, percentage with only a high school diploma, unemployment rate, poverty rate, and

Table III: Non-tradable employment growth and the housing net worth shock.

	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A (OLS): Effect of change in housing net worth on non-tradable employment growth by establishment size (N = 944 counties)</i>						
	Establishment Size in Terms of Number of Employees:					
	1 to 4	5 to 9	10 to 19	20 to 49	50 to 99	100+
Δ Housing Net Worth, 2006–2009	0.070** (0.025)	0.032 (0.036)	0.022 (0.044)	0.134** (0.032)	0.152 (0.097)	0.434** (0.061)
<i>Panel B (IV): Effect of change in housing net worth on non-tradable employment growth by establishment size (N = 540 counties)</i>						
	Establishment Size in Terms of Number of Employees:					
	1 to 4	1 to 4	1 to 4	1 to 4	1 to 4	1 to 4
Δ Housing Net Worth, 2006–2009	–0.134 (0.147)	0.000 (0.125)	–0.022 (0.109)	0.193* (0.086)	0.335 (0.191)	0.770** (0.208)
<i>Panel C: Effect of change in housing net worth on non-tradable employment growth by banking type</i>						
	Banking Type:					
	National (OLS, N = 472)	Local (OLS, N = 304)	National (IV, N = 472)	Local (IV, N = 236)		
Δ Housing Net Worth, 2006–2009	0.186** (0.041)	0.306 (0.178)	0.233** (0.068)	0.308** (0.107)		

*This table reports regression estimates of non-tradable employment growth from 2007 to 2009 on the housing net worth shock at the county level. Panels A and B reports the OLS and IV coefficient estimates, respectively, for establishments of varying sizes. Panel C reports the coefficients separately for national and local banking markets. Non-tradable employment is defined as employment in restaurant and retail industries at the four-digit industry level and then aggregated up separately for each county. All regressions are weighted using the total number of households in a county as weights. The instrumental variables specifications use the housing supply elasticity as an instrument for the change in housing net worth in the first stage. Standard errors are clustered at the state level. **, * Coefficient statistically different from zero at the 1% and 5% confidence level, respectively

Table IV: Credit-supply tightening tests.

6.4 Figure 2: Labor-market adjustment margins

Read:

- The top panels show that tradable employment does not rise in counties with larger housing net worth losses.
- The middle panels show weak wage adjustment. Payroll wage growth has a mild relationship, but hourly wage growth does not show a strong compensating decline.
- The bottom panels show little evidence that population or labor force growth adjusts enough to undo the local shock.
- Overall, the figure rejects the simple flexible-adjustment story in which local non-tradable losses are offset by tradable expansion, wage declines, or migration.

Economic message: local demand shocks translated into job losses because labor markets did not reallocate quickly enough across sectors or locations.

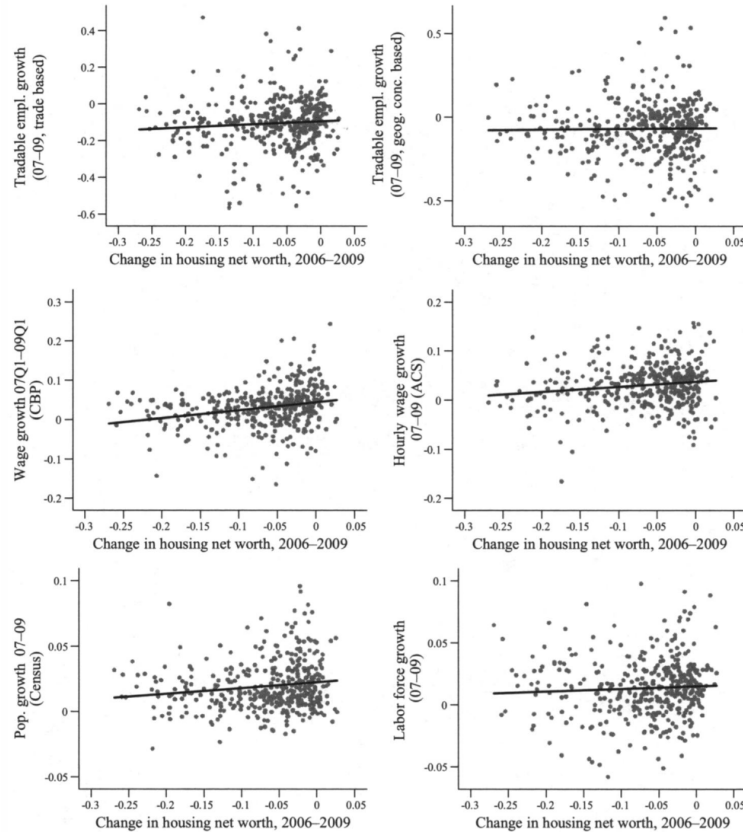


Figure 2: Tradable employment, wages, labor mobility, and the housing net worth shock.

6.5 Tables V and VI: Tradable employment, wages, and mobility

Read Table V:

- Tradable employment growth is not significantly related to the housing net worth shock in the county-level specifications.
- In county-industry specifications, the housing net worth effect weakens as industries become more geographically concentrated.
- This supports the interpretation that the shock works through local demand rather than a broad county productivity shock.

Read Table VI:

- Wage measures do not show large downward adjustment in harder-hit counties.
- Population growth, in-migration, and labor force growth do not show robust adjustment patterns that would offset the shock.
- The evidence is consistent with nominal wage rigidity, real frictions, or slow mobility.

Economic message: the absence of tradable-sector expansion and the weak wage/mobility responses

explain why local housing net worth losses can produce persistent employment losses.

TABLE V
TRADEABLE EMPLOYMENT GROWTH AND THE HOUSING NET WORTH SHOCK*

Tradable Definition Used:	Employment Growth, 2007–2009 (County–Four Digit Industry Level)							
	Employment Growth, Tradable Industries, 2007–2009				All Industries			
	Global Trade (1)	Geog. Conc. (2)	Global Trade (3)	Geog. Conc. (4)	Geog. Conc. (5)	Geog. Conc. (6)	Global Trade (7)	Global Trade (8)
Δ Housing Net Worth, 2006–2009	0.018 (0.099)	–0.085 (0.063)	0.064 (0.098)	–0.063 (0.074)	0.198** (0.058)	—	–0.096 (0.137)	—
Industry Geographical Herfindahl Index					–2.936** (0.606)	—		
Δ HNW* (Geographical Herfindahl)					–11.733** (3.254)	–7.328** (1.367)		
Trade per worker (Mil. \$)							–0.262 (0.238)	—
Δ HNW* (Trade per worker)							–1.274 (1.386)	–1.409 (0.980)
Constant	–0.114** (0.012)	–0.091** (0.012)	–0.286 (0.950)	0.542 (1.144)	–0.077** (0.010)	—	–0.233** (0.015)	—
Two-digit 2006 employment share controls?			Yes	Yes			Yes	Yes
Four-digit industry fixed effects							Yes	Yes
County Fixed Effects							Yes	Yes
N	944	944	944	944	180,756	180,756	31,970	31,970
R ²	0.000	0.002	0.079	0.064	0.004	0.18	0.0014	0.19

*This table reports regression estimates of tradable employment growth from 2007 to 2009 on the housing net worth shock at the county level. Tradable employment is defined at the four-digit industry level and then aggregated up separately for each county in columns 1 through 4. We use two different definitions of tradable industries, one based on US imports and exports, and another based on an industry's geographical concentration within the United States. There are 23 two-digit industry employment share variables as controls in columns 5 and 6. A unit of observation in columns 5 through 8 is county–four digit industry. There are 294 four digit industry and 944 county fixed effects in columns 6 and 8. All regressions are weighted using the total number of households as weights. Standard errors are clustered at the state level. **, * Coefficient statistically different than zero at the 1% and 5% confidence level, respectively.

Table V: Tradable employment growth and the housing net worth shock.

7 Short summary

- The paper studies whether household balance sheet deterioration explains employment losses during 2007–2009.
- The key shock is the decline in housing net worth at the county level.
- The key empirical split is between non-tradable jobs, which depend on local demand, and tradable jobs, which depend on national or global demand.
- Counties with larger housing net worth losses experience larger declines in non-tradable employment.
- The result holds under two independent industry-classification schemes.
- The relationship is not explained by construction exposure or by differences in broad industry composition.
- IV estimates using housing supply elasticity support the main result.
- Credit-supply tightening is unlikely to be the main mechanism because the effect is stronger among larger establishments and appears in both national- and local-banking counties.
- Tradable employment does not expand in the counties with the largest housing net worth losses.
- Wages and migration also do not adjust strongly enough to offset the local demand shock.
- The main takeaway is that local household balance sheet deterioration reduced local demand and caused real employment losses because adjustment margins were weak.

TABLE VI
WAGES, MOBILITY, AND THE HOUSING NET WORTH SHOCK*

	Total Wage Growth, 2007 to 2009, CBP		Average Hourly Wage Growth, 2007 to 2009, ACS		Population Growth, 2007–2009		In-Migration Growth, 2007–2009		Labor Force Growth, 2007–2009	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Δ Housing Net Worth, 2006–2009	0.061 (0.041)	0.078* (0.037)	0.054 (0.039)	0.056 (0.035)	0.019 (0.021)	0.057** (0.021)	–0.042 (0.11)	–0.128 (0.127)	–0.0094 (0.020)	0.0032 (0.024)
Constant	0.031** (0.007)	–0.325 (0.250)	0.037** (0.003)	0.078 (0.20)	0.021** (0.004)	–0.103 (0.137)	–0.010** (0.015)	–0.530 (1.778)	0.0136 (0.004)	0.030 (0.24)
Two-digit 2006 employment share controls?		Yes		Yes		Yes		Yes		Yes
N	944	944	943	943	939	939	943	943	944	944
R ²	0.012	0.16	0.018	0.076	0.009	0.25	0	0.027	0.001	0.12

*Columns 1 through 4 present coefficients from regressions relating wage growth in a county from 2007 to 2009 to the housing net worth shock. The specifications in columns 1 and 2 use total wages from the Census County Business Patterns data, and columns 3 and 4 use hourly wage growth data from the American Community Survey. Columns 5 through 10 present coefficients from regressions relating mobility and labor force participation in a county from 2007 to 2009 to the change in housing net worth. Columns 5 and 6 use census data on population growth, columns 7 and 8 use growth in in-migration from the American Community Survey, and columns 9 and 10 use labor force growth from the Bureau of Labor Statistics. All regressions are weighted using the total number of households in a county as weights. Standard errors are clustered at the state level. **, * Coefficient statistically different than zero at the 1% and 5% confidence level, respectively.

Table VI: Wages, mobility, and the housing net worth shock.

8 Conclusion

8.1 Core conclusion

Mian and Sufi show that the collapse in household housing net worth was an important driver of the 2007–2009 employment decline. Counties with larger housing net worth losses experienced larger declines in non-tradable employment, exactly where a local demand channel should show up.

8.2 Economic intuition

A fall in house prices reduces household net worth and collateral value. Households then reduce spending. Restaurants, retail stores, and other local-service industries lose customers and cut employment. In a frictionless economy, wages would fall and workers would move into tradable industries or other counties. The paper finds little evidence that this adjustment happened quickly.

8.3 Incremental contribution revisited

The paper moves the household-balance-sheet literature from consumption to employment. It shows that spending declines caused by housing net worth losses translated into local job losses, and it uses the tradable/non-tradable distinction to separate local demand effects from national shocks.

8.4 Implications

The findings imply that recessions caused by household balance sheet shocks can be severe when nominal and real adjustment margins are limited. Stabilizing household balance sheets, or preventing sharp collapses in collateral values, may matter for employment because local demand losses can be persistent.

8.5 Limitations

The empirical design uses cross-county variation and therefore identifies relative local effects most directly. Aggregate effects require additional general-equilibrium assumptions, because local demand shocks can spill across counties through trade and migration. The housing net worth shock may also be correlated with other local conditions, though the paper performs many robustness checks.

8.6 Takeaway

The enduring takeaway is that the Great Recession was not only a financial-sector event or a construction-sector bust. It was also a household balance sheet recession: when housing net worth collapsed, local demand collapsed, and local non-tradable employment fell sharply in the hardest-hit counties.